THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 60°	The angle and its complement add up to 90°.
2	**(a)** 70°	Find the supplement first, then halve it.
3	**(c)** 60°	They add up to 180°.
4	**(c)** 90°	The angles around a point add up to 360°.
5	**(b)** 75°	Let the angle be y; its supplement is $y + a$.
6	**(a)** parallel	Check the angle relation.
7	**(a)** 100°	Larger + smaller $= 180^\circ$ and larger − smaller $= d$.
8	**(c)** 60°	Share 90° in the given ratio.
9	**(c)** 75°	The angle and its complement add up to 90°.
10	**(a)** 60°	Find the supplement first, then halve it.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let the complement be y; the angle is $y + 20$ and $y + (y + 20) = 90$; $2y = 70$, so $y = 35$ (1); The angle is 55° (1).	1+1
12	Larger + smaller $= 180$ and larger − smaller $= 20$; Larger $= (180 + 20) \div 2 = 100$ (1); The angles are 100° and 80° (1).	1+1
13	Compare 80° and 80° (1); Equal alternate interior angles, so the lines are parallel (1).	1+1
14	Let the complement be y; the angle is $y + 40$ and $y + (y + 40) = 90$; $2y = 50$, so $y = 25$ (1); The angle is 65° (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Adjacent angles form a linear pair: 5 parts $= 180^\circ$ (1); One part $= 36^\circ$, so the adjacent angles are 36° and 144° (1); Vertically opposite angles are equal, so the four angles are 36°, 144°, 36°, 144° (1).	1+1+1
16	Let the complement be y; the angle is $2y + 6$ and $y + (2y + 6) = 90$ (1); $3y = 84$, so $y = 28$ (1); The angle is 62° and its complement is 28° (1).	1+1+1
17	4 parts $= 180^\circ$, one part $= 45^\circ$ (1); The angles are 45° and 135° (1); Complement of the smaller angle $= 90^\circ - 45^\circ = 45^\circ$ (1).	1+1+1
	OR: Adjacent angles form a linear pair: 9 parts $= 180^\circ$ (1); One part $= 20^\circ$, so the adjacent angles are 20° and 160° (1); Vertically opposite angles are equal, so the four angles are 20°, 160°, 20°, 160° (1).	1+1+1
18	Let the complement be y; the angle is $2y + 15$ and $y + (2y + 15) = 90$ (1); $3y = 75$, so $y = 25$ (1); The angle is 65° and its complement is 25° (1).	1+1+1
	OR: 9 parts $= 180^\circ$, one part $= 20^\circ$ (1); The angles are 40° and 140° (1); Complement of the smaller angle $= 90^\circ - 40^\circ = 50^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	They are supplementary: 5 parts $= 180^\circ$ (1); One part $= 36^\circ$, so the angles are 72° and 108° (2); Linear pair with the larger: $180^\circ - 108^\circ = 72^\circ$ (2).	1+2+2
	OR: Total parts $= 10$ and $360^\circ \div 10 = 36^\circ$ (1); The angles are 36°, 72°, 108°, 144° (2); Kinds: 36° acute, 72° acute, 108° obtuse, 144° obtuse (2).	1+2+2
20	They are supplementary: 6 parts $= 180^\circ$ (1); One part $= 30^\circ$, so the angles are 30° and 150° (2); Linear pair with the larger: $180^\circ - 150^\circ = 30^\circ$ (2).	1+2+2
	OR: Total parts $= 15$ and $360^\circ \div 15 = 24^\circ$ (1); The angles are 48°, 72°, 96°, 144° (2); Kinds: 48° acute, 72° acute, 96° obtuse, 144° obtuse (2).	1+2+2

Mistakes to avoid:

- Mixing up complementary (90°) and supplementary (180°) angles.
- Saying vertically opposite angles are supplementary; they are equal.
- Saying corresponding or alternate interior angles are supplementary; same-side interior angles are the supplementary pair.
- Using the corresponding angle result when the lines are not known to be parallel.
- Forgetting that the four angles around a point add up to 360°, not 180°.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 72°	They add up to 180°.
2	**(c)** 85°	The angles around a point add up to 360°.
3	**(c)** 65°	Let the angle be y; its supplement is $y + a$.
4	**(b)** not parallel	Check the angle relation.
5	**(d)** 110°	Larger + smaller $= 180^\circ$ and larger − smaller $= d$.
6	**(c)** 54°	Share 90° in the given ratio.
7	**(c)** 80°	The angle and its complement add up to 90°.
8	**(d)** 40°	Find the supplement first, then halve it.
9	**(d)** 30°	They add up to 180°.
10	**(a)** 75°	The angles around a point add up to 360°.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Larger + smaller $= 180$ and larger − smaller $= 40$; Larger $= (180 + 40) \div 2 = 110$ (1); The angles are 110° and 70° (1).	1+1
12	Compare 115° and 125° (1); The alternate interior angles are not equal, so the lines are not parallel (1).	1+1
13	Let the complement be y; the angle is $y + 60$ and $y + (y + 60) = 90$; $2y = 30$, so $y = 15$ (1); The angle is 75° (1).	1+1
14	Larger + smaller $= 180$ and larger − smaller $= 60$; Larger $= (180 + 60) \div 2 = 120$ (1); The angles are 120° and 60° (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Adjacent angles form a linear pair: 10 parts $= 180^\circ$ (1); One part $= 18^\circ$, so the adjacent angles are 18° and 162° (1); Vertically opposite angles are equal, so the four angles are 18°, 162°, 18°, 162° (1).	1+1+1
16	Let the complement be y; the angle is $2y + 30$ and $y + (2y + 30) = 90$ (1); $3y = 60$, so $y = 20$ (1); The angle is 70° and its complement is 20° (1).	1+1+1
17	18 parts $= 180^\circ$, one part $= 10^\circ$ (1); The angles are 10° and 170° (1); Complement of the smaller angle $= 90^\circ - 10^\circ = 80^\circ$ (1).	1+1+1
	OR: Adjacent angles form a linear pair: 15 parts $= 180^\circ$ (1); One part $= 12^\circ$, so the adjacent angles are 24° and 156° (1); Vertically opposite angles are equal, so the four angles are 24°, 156°, 24°, 156° (1).	1+1+1
18	Let the complement be y; the angle is $2y + 45$ and $y + (2y + 45) = 90$ (1); $3y = 45$, so $y = 15$ (1); The angle is 75° and its complement is 15° (1).	1+1+1
	OR: 15 parts $= 180^\circ$, one part $= 12^\circ$ (1); The angles are 48° and 132° (1); Complement of the smaller angle $= 90^\circ - 48^\circ = 42^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	They are supplementary: 9 parts $= 180^\circ$ (1); One part $= 20^\circ$, so the angles are 80° and 100° (2); Linear pair with the larger: $180^\circ - 100^\circ = 80^\circ$ (2).	1+2+2
	OR: Total parts $= 12$ and $360^\circ \div 12 = 30^\circ$ (1); The angles are 30°, 60°, 120°, 150° (2); Kinds: 30° acute, 60° acute, 120° obtuse, 150° obtuse (2).	1+2+2
20	They are supplementary: 12 parts $= 180^\circ$ (1); One part $= 15^\circ$, so the angles are 15° and 165° (2); Linear pair with the larger: $180^\circ - 165^\circ = 15^\circ$ (2).	1+2+2
	OR: Total parts $= 12$ and $360^\circ \div 12 = 30^\circ$ (1); The angles are 30°, 90°, 120°, 120° (2); Kinds: 30° acute, 90° right, 120° obtuse, 120° obtuse (2).	1+2+2

Mistakes to avoid:

- Mixing up complementary (90°) and supplementary (180°) angles.
- Saying vertically opposite angles are supplementary; they are equal.
- Saying corresponding or alternate interior angles are supplementary; same-side interior angles are the supplementary pair.
- Using the corresponding angle result when the lines are not known to be parallel.
- Forgetting that the four angles around a point add up to 360°, not 180°.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Parallel and Intersecting Lines** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 50°	Let the angle be y; its supplement is $y + a$.
2	**(b)** parallel	Check the angle relation.
3	**(a)** 120°	Larger + smaller $= 180^\circ$ and larger − smaller $= d$.
4	**(c)** 50°	Share 90° in the given ratio.
5	**(b)** 85°	The angle and its complement add up to 90°.
6	**(d)** 30°	Find the supplement first, then halve it.
7	**(a)** 80°	They add up to 180°.
8	**(d)** 55°	The angles around a point add up to 360°.
9	**(a)** 30°	Let the angle be y; its supplement is $y + a$.
10	**(d)** not possible	Check the angle relation.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Compare 47° and 47° (1); Equal alternate interior angles, so the lines are parallel (1).	1+1
12	Let the complement be y; the angle is $y + 10$ and $y + (y + 10) = 90$; $2y = 80$, so $y = 40$ (1); The angle is 50° (1).	1+1
13	Larger + smaller $= 180$ and larger − smaller $= 100$; Larger $= (180 + 100) \div 2 = 140$ (1); The angles are 140° and 40° (1).	1+1
14	Compare 92° and 88° (1); The alternate interior angles are not equal, so the lines are not parallel (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Adjacent angles form a linear pair: 12 parts $= 180^\circ$ (1); One part $= 15^\circ$, so the adjacent angles are 105° and 75° (1); Vertically opposite angles are equal, so the four angles are 105°, 75°, 105°, 75° (1).	1+1+1
16	Let the complement be y; the angle is $2y + 12$ and $y + (2y + 12) = 90$ (1); $3y = 78$, so $y = 26$ (1); The angle is 64° and its complement is 26° (1).	1+1+1
17	12 parts $= 180^\circ$, one part $= 15^\circ$ (1); The angles are 75° and 105° (1); Complement of the smaller angle $= 90^\circ - 75^\circ = 15^\circ$ (1).	1+1+1
	OR: Adjacent angles form a linear pair: 15 parts $= 180^\circ$ (1); One part $= 12^\circ$, so the adjacent angles are 12° and 168° (1); Vertically opposite angles are equal, so the four angles are 12°, 168°, 12°, 168° (1).	1+1+1
18	Let the complement be y; the angle is $2y + 24$ and $y + (2y + 24) = 90$ (1); $3y = 66$, so $y = 22$ (1); The angle is 68° and its complement is 22° (1).	1+1+1
	OR: 20 parts $= 180^\circ$, one part $= 9^\circ$ (1); The angles are 9° and 171° (1); Complement of the smaller angle $= 90^\circ - 9^\circ = 81^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	They are supplementary: 10 parts $= 180^\circ$ (1); One part $= 18^\circ$, so the angles are 54° and 126° (2); Linear pair with the larger: $180^\circ - 126^\circ = 54^\circ$ (2).	1+2+2
	OR: Total parts $= 12$ and $360^\circ \div 12 = 30^\circ$ (1); The angles are 60°, 60°, 90°, 150° (2); Kinds: 60° acute, 60° acute, 90° right, 150° obtuse (2).	1+2+2
20	They are supplementary: 3 parts $= 180^\circ$ (1); One part $= 60^\circ$, so the angles are 60° and 120° (2); Linear pair with the larger: $180^\circ - 120^\circ = 60^\circ$ (2).	1+2+2
	OR: Total parts $= 12$ and $360^\circ \div 12 = 30^\circ$ (1); The angles are 30°, 60°, 90°, 180° (2); Kinds: 30° acute, 60° acute, 90° right, 180° straight (2).	1+2+2

Mistakes to avoid:

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- Saying vertically opposite angles are supplementary; they are equal.
- Saying corresponding or alternate interior angles are supplementary; same-side interior angles are the supplementary pair.
- Using the corresponding angle result when the lines are not known to be parallel.
- Forgetting that the four angles around a point add up to 360°, not 180°.

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